

SavePredictedExams — SME

Economics

Higher level

Paper 3

SME Predicted Paper Set — Moderate

1 hour

Instructions to candidates

- Do not open this examination paper until instructed to do so.
- You are permitted access to a calculator for this paper.
- Answer **all** questions.
- Answers must be written within the answer boxes provided.
- Unless otherwise stated in the question, all numerical answers must be given exactly or correct to two decimal places.
- You must show all your working.
- Use fully labelled diagrams where appropriate.
- The maximum mark for this examination paper is **[60 marks]**.

1. Palm oil is the most widely consumed vegetable oil in the world, used in foods, cosmetics and biofuels. Indonesia is the largest producer, accounting for roughly 59% of global supply in 2022. Large-scale palm oil cultivation has been linked to deforestation, biodiversity loss and greenhouse-gas emissions, generating **negative externalities of production**.

(a) (i) Define the term **externality**. [2]

Figure 1 below shows the market for palm oil in Indonesia. The marginal private cost (MPC) lies below the marginal social cost (MSC). The market equilibrium price is \$900 per tonne at a quantity of 60 million tonnes. The socially optimal price is \$1,200 per tonne at 45 million tonnes. Demand represents both MPB and MSB.

Variable	Value
Private market price (P_e)	\$900 per tonne
Private market quantity (Q_e)	60 million tonnes
Socially optimal price (P_o)	\$1,200 per tonne
Socially optimal quantity (Q_o)	45 million tonnes
Vertical gap between MSC and MPC at Q_e	\$500 per tonne

(ii) Using Figure 1, identify the value of the external cost per tonne of palm oil at the free-market equilibrium. [1]

(iii) Using Figure 1, calculate the welfare loss arising from the overproduction of palm oil. [2]

(iv) Explain, using a marginal-cost/marginal-benefit diagram, why negative externalities of production lead to an allocatively inefficient outcome. [4]

(v) State **one** characteristic of a common-pool resource. [1]

(vi) Outline the meaning of the term **tragedy of the commons**. [2]

The Indonesian government is considering imposing a **Pigouvian tax** of \$500 per tonne on palm-oil producers.

(vii) Calculate the expected tax revenue if, after the tax is imposed, 45 million tonnes of palm oil are produced. [2]

(viii) Explain, using a diagram, how a Pigouvian tax can internalise the negative externality associated with palm-oil production. [4]

Research suggests the price elasticity of demand (PED) for palm oil from EU buyers is -0.45 .

(ix) Using the information above, state whether palm oil is price elastic or price inelastic and give one reason for your answer. [2]

(b) Using the text/data provided and your knowledge of economics, **recommend** a policy that the Indonesian government could introduce to reduce the environmental damage caused by palm-oil production. [10]

2. Ethiopia is one of the fastest-growing economies in sub-Saharan Africa. Despite strong growth, the country continues to face **high inflation**, a widening **current account deficit** and rising **unemployment** among youth. Table 1 below provides selected economic data for Ethiopia.

	Nominal GDP (billion USD)	GDP deflator (base = 100)	Population (million)
2022	126.80	115.0	123.4
2023	155.80	140.3	126.5

(a) (i) Define the term **real GDP**. [2]

(ii) Using Table 1, calculate Ethiopia's real GDP in 2023. [2]

(iii) Using Table 1 and your answer to (a)(ii), calculate Ethiopia's real GDP per capita in 2023. [2]

(iv) State **two** limitations of using real GDP per capita as a measure of living standards. [2]

(v) Using the information in Table 1, calculate the rate of inflation in Ethiopia between 2022 and 2023. [2]

(vi) Outline **one** cost of high inflation to an economy. [2]

(vii) Explain, using an AD/AS diagram, how a rise in the world price of oil could lead to cost-push inflation in Ethiopia. [4]

In 2023 Ethiopia had a labour force of 54 million workers and 4.86 million were unemployed.

(viii) Calculate the unemployment rate in Ethiopia in 2023. [2]

(ix) Distinguish between **cyclical** and **structural** unemployment. [2]

(b) Using the text/data provided and your knowledge of economics, **recommend** a policy that the Ethiopian government could introduce to reduce unemployment. [10]